

Supplementary Materials for AI-Empowered Cooperative One-Surgeon-Four-Arm Robotic System for Solo-Hysterectomy

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This PDF file includes:

Materials and Methods

Figures S1 to S4

Tables S1 to S3

Captions for Movies S1 to S9

Other Supplementary Materials for this manuscript:

Movies S1 to S9

Materials and Methods

FoV control strategy

The error update for variable x in the control process is calculated as follows:

$$e(x) = \begin{cases} 0, & \text{if } (\neg F \wedge x \in [\sigma_{min}^l, \sigma_{max}^l]) \vee (F \wedge x \in [\sigma_{min}^s, \sigma_{max}^s]) \\ x - \sigma_{min}^s, & \text{if } F \wedge x < \sigma_{min}^s \\ x - \sigma_{max}^s, & \text{if } F \wedge x > \sigma_{max}^s \end{cases} \quad (S1)$$

$$F = \begin{cases} \text{whether } x \in [\sigma_{min}^s, \sigma_{max}^s], & \text{if } F \text{ is True} \\ \text{whether } x \in [\sigma_{min}^l, \sigma_{max}^l], & \text{others} \end{cases}$$

where variable F represents whether the control process of the dynamic scale is activated or not. In addition, considering the inclination of surgeons to switch to faster laparoscopic movement and reposition the view to an off-center region when moving the instrument to the edge of the image, we adopt a nonlinear fitting strategy to dynamically generate an intermediate region with scale s based on the current instrument deviation $p_{x,y}$ to achieve this effect:

$$\begin{aligned} s_{x,y}(p) &= c_1 \cdot \exp\left(\frac{p-p_{min}}{p_{max}-p_{min}}\right) - c_2 \\ s(p) &= \min(s_x, s_y) \end{aligned} \quad (S2)$$

where coefficient $c_1 = \ln 2$, $c_2 = 1$. The upper and lower bounds of the internal region $\sigma_{min/max}^s$ corresponding to the xy -axes of the image are $(1 \pm s) \cdot \frac{w}{2}$ and $(1 \pm s) \cdot \frac{h}{2}$, respectively.

Regulatory approval for the cadaver study

The experiment was conducted on a human cadaveric specimen at the CUHK Jockey Club Minimally Invasive Surgical Skills Centre, with the specimen obtained and used through approved institutional procedures.

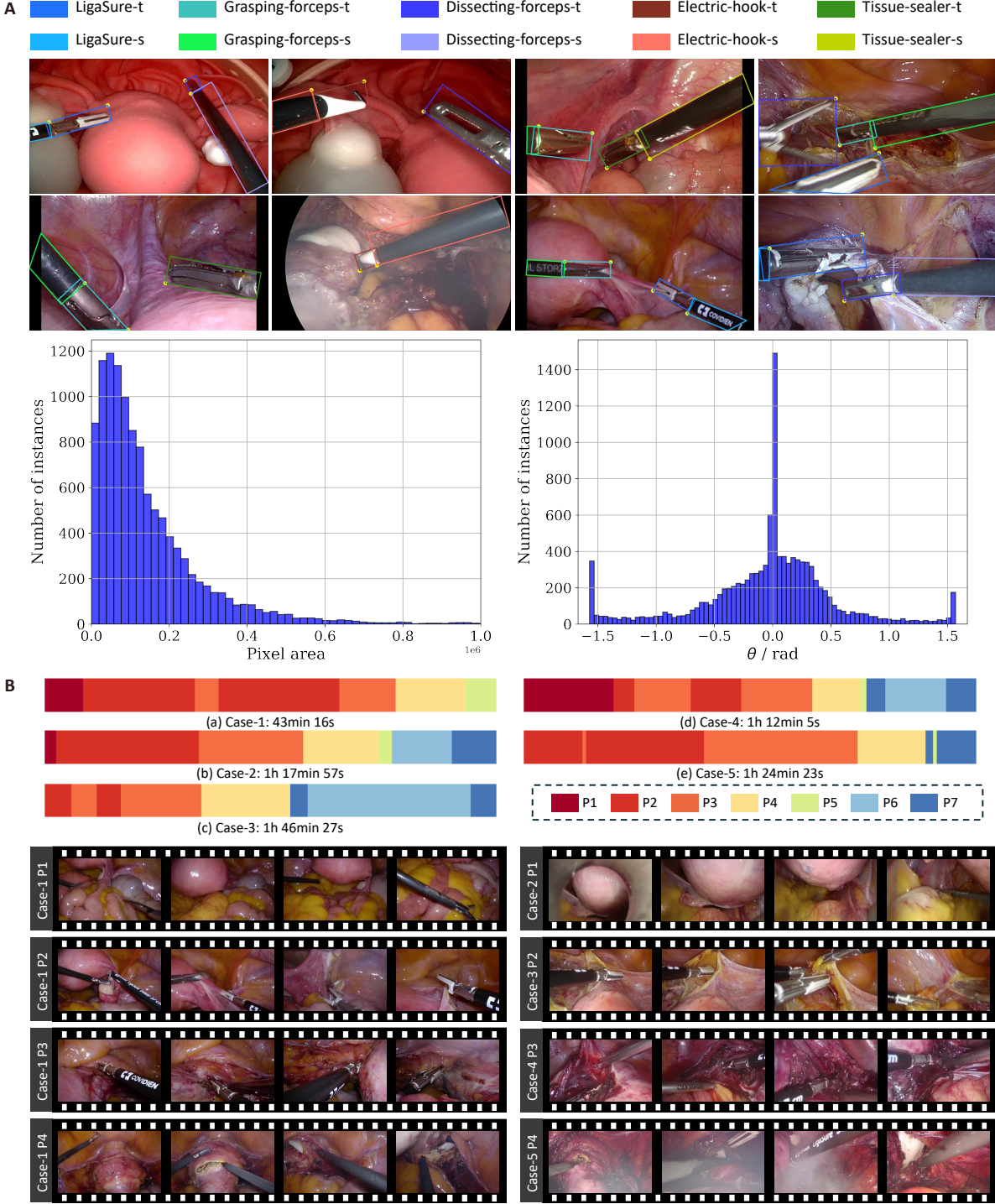


Figure S1: Dataset description. Typical images of ten object categories containing the tip(-t) and shaft(-s) of five instruments in the self-collected dataset. **The statistics of the self-collected dataset.** (a) The number of instances and their pixel area. (b) The number of instances and their rotated angle θ . (c) The distribution of the pixel area for each type of instrument.

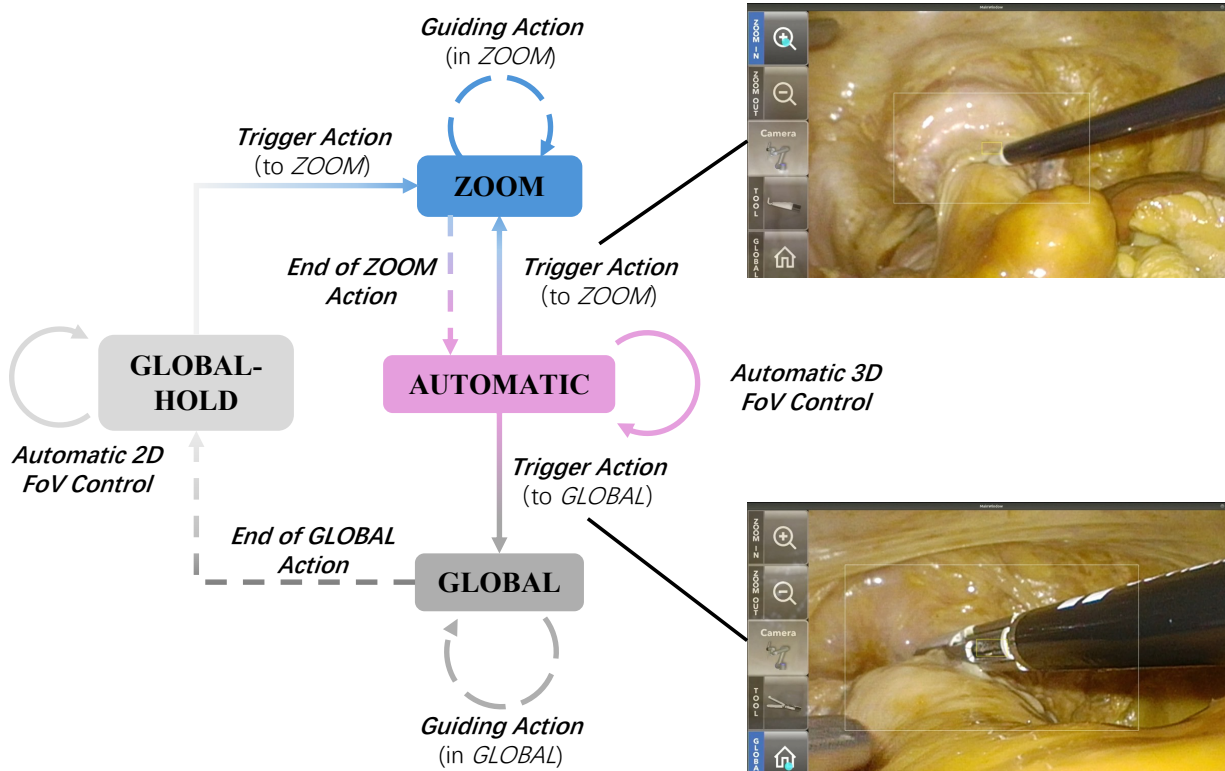


Figure S2: State transition diagram for semi-autonomous laparoscope control. The laparoscope is controlled in *AUTOMATIC* mode most of the time, and the surgeon can trigger the action into the *ZOOM* or *GLOBAL* state if necessary.

The instrument's position in the image is considered as the mouse pointer, and button clicking and releasing are triggered based on the instrument's presence in the button area for a certain duration. The process for controlling the motion state of the laparoscope by moving instruments is outlined as follows:

- *Switching from AUTOMATIC state to ZOOM state:* To complete the transition, the surgeon can activate the laparoscope panel and move the instrument to click the *Zoom-in/out* button in the image. Note that the state will back to the *AUTOMATIC* state when the zoom-in/out action is finished, and then the laparoscope will actively track the instrument with the current distance interval.
- *Switching from AUTOMATIC state to GLOBAL state:* Similar to Step 1, the surgeon can virtually click the *Global* button to return to the global position automatically. Note that the

state will switch to *GLOBAL-HOLD* state when the global action is finished.

- *Switching from GLOBAL-HOLD state to AUTOMATIC state:* As there is no depth control in *GLOBAL-HOLD* state, and the surgeon can activate the depth control by virtually clicking the *Zoom-in/out* button, where the system will first enter the *ZOOM* state and then return to the *AUTOMATIC* state when the zoom action is finished.

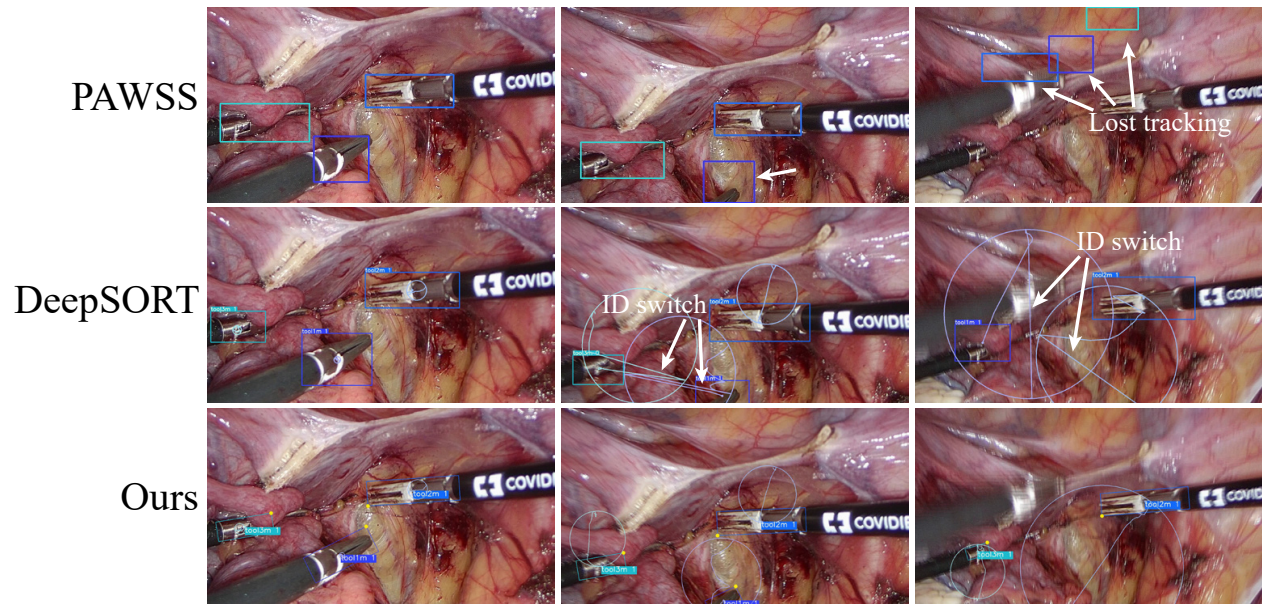


Figure S3: The tracking results on our in-house video frames. The different colors represent the different tip regions of the instrument, and the yellow point in our results indicates the tracked upper left index. The trajectory represents the historical position over the last T timestamps, and a minimum enclosing circle is used to show the moving range of the instrument.

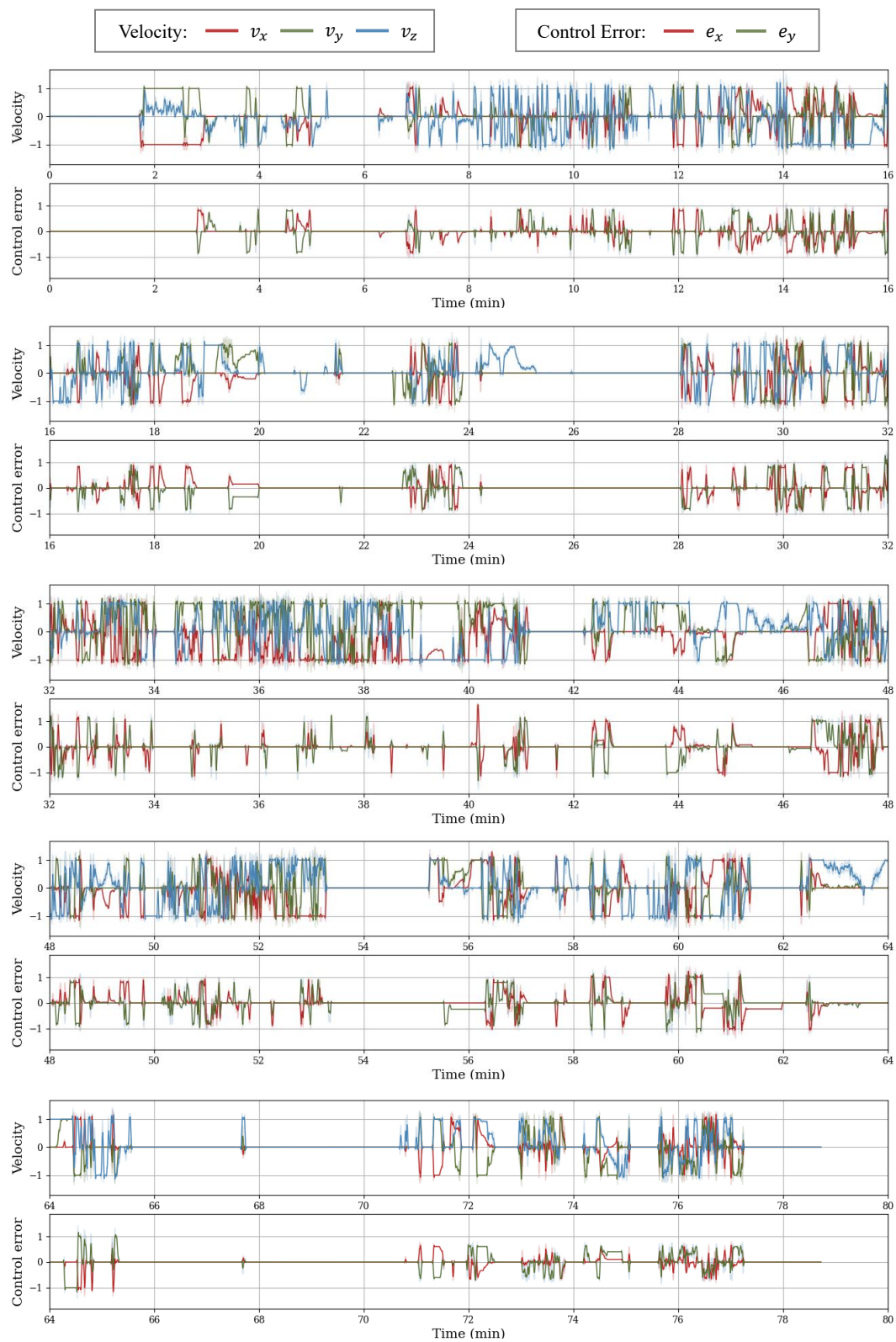


Figure S4: The control velocity and control errors for laparoscope manipulation throughout the entire cadaver study.

Table S1: The triggered action description of each button in the laparoscopic control panel.

Panel	Button	Action description
Camera	<i>Zoom in/out</i>	The laparoscope is first inserted/retracted a certain distance to magnify/reduce the FoV. Then its 6D pose was adjusted to maintain the modified zoom level to track the main instrument.
	<i>Main instrument</i>	A prompt box superimposed on the current image will pop up and the surgeon can place a new instrument in the box and maintain it for a certain amount of time, switching it to the instrument to be tracked for automatic FoV adjustment.
	<i>Global view</i>	The laparoscope first returns to the position corresponding to the global view, and is then controlled automatically to track the main instrument in the 2D image plane without the zoom-related control.
Uterus	<i>Up/Down</i>	The uterus is lifted/lowered at a certain speed to expose the area of interest for manipulation.
	<i>Left/Right</i>	Turn the uterus left/right at a certain speed to expose the area of interest for manipulation.
	<i>Insert/Retract</i>	The uterus is inserted/retracted at a certain speed to control tissue tension.

Table S2: Quantitative comparison of the instrument tracking performance under various conditions using PAWSS, DeepSORT, and our proposed MOIT-Net + LTMT framework.

Experiments		mIoU	mMOTA	IDs Box/UL	IDP Box/UL	mAcc
PAWSS	<i>Exp1</i>	58.0	42.5	-	-	-
	<i>Exp2</i>	46.3	34.4	-	-	-
	<i>Exp3</i>	45.8	34.8	-	-	-
	<i>Exp-L</i>	53.4	3.8	-	-	-
DeepSORT	<i>Exp1</i>	82.8	84.5	9 / -	94.8 / -	87.2
	<i>Exp2</i>	75.7	66.8	16 / -	92.3 / -	71.7
	<i>Exp3</i>	76.8	75.3	38 / -	88.9 / -	82.6
	<i>Exp-L</i>	84.3	85.9	294 / -	95.2 / -	89.1
Ours	<i>Exp1</i>	90.5	96.7	0 / 0	100 / 100	97.4
	<i>Exp2</i>	87.9	96.4	1 / 0	99.5 / 100	96.8
	<i>Exp3</i>	86.8	94.9	0 / 3	100 / 92.5	95.5
	<i>Exp-L</i>	92.4	98.2	4 / 6	99.6 / 99.2	98.6

We employed the following metrics to quantify the tracking results: (1) The mean IoU (mIoU) is used to measure the accuracy of instrument localization, (2) The mean accuracy (mAcc) is used to measure the accuracy of the tracked box and the upper-left index, (3) The mean multiple object tracking accuracy (mMOTA) is used to measure the continuous multiple instrument tracking performance, (4) The total counts of the identity switches (IDs) and the identity precision (IDP) are used to measure the number of interruptions in the tracking process and the accuracy of the tracked instrument, which is related to the security of subsequent interactions.

We first conducted the experiments on clinical surgical video clips of around 1 minute under typical conditions, including those where FoV and instrument motions were relatively slow (*Exp1*) or fast (*Exp2*), and the dynamic scenarios with smoke and blood contamination (*Exp3*). Then we tested a 26-minute video including the aforementioned various complex scenarios (*Exp4*) to evaluate the overall long-term tracking performance.

Table S3: List of Acronyms.

Acronym	Full Name
AI	Artificial Intelligence
CSL	Circle Smooth Label
DoF	Degree of Freedom
ETC	Equilateral Triangle Configuration
FK	Forward Kinematics
FoV	Field of View
GT	Ground-Truth
GUI	Graphical User Interface
HMI	Human-Machine Interface
IMP	Instrument Mouse Perception
LMC	Laparoscopic Manipulator Control
LoA	Level of Autonomy
LoCC	Level of Cooperative Control
LTMT	Long-Term Memory-based Tracker
MIS	Minimally Invasive Surgery
MTOI-Net	Multi-Task Orientated Instrument Network
NLS	Nature Line-of-Sight
OGI	Overlay Graphical Interface
OSFA	One-Surgeon-For-Arms
RCM	Remote Center of Motion
SMBG	Single-Motor Bilinear-Guided
STRUM	inSTRUment Mouse
TLH	Total Laparoscopic Hysterectomy
UMC	Uterine Manipulator Control
ZVF	Z-axis Virtual Fixture

Caption for Movie S1. Experimental results of surgical instrument detection and tracking.

It demonstrates robust real-time tracking of multiple instruments in real-world surgical videos, addressing various complex scenarios such as smoke, blur, occlusion, blood, and small objects, etc.

Caption for Movie S2. Experimental results of surgical phase recognition. It showcases different laparoscope control strategies, including optimal view selection and control speed, tailored to the current surgical phase.

Caption for Movie S3. Phantom experiments on LoCC-0. The experiments were conducted by one surgeon and two assistants, with manual operation of the laparoscope and the uterine manipulation rod.

Caption for Movie S4. Phantom experiments on LoCC-1. The laparoscope-holding robot arm and the robotic uterine manipulator are developed to replace the assistant, and the assistants are responsible for the tele-operation control of the two robots.

Caption for Movie S5. Phantom experiments on LoCC-2. Two robots are directly controlled by the primary surgeon through external input commands.

Caption for Movie S6. Phantom experiments on LoCC-3. The movements of two robots are determined by AI-based surgical scene perception algorithms, enabling autonomous control without assistant and extra equipment.

Caption for Movie S7. Phantom experiments on LoCC-4. The primary surgeon uses our developed GUI to directly control the robots to manipulate the laparoscope and uterus to the desired position.

Caption for Movie S8. Cadaver experiment: Intelligent control for the robotic laparoscope and uterus manipulator. We individually validated the performance of each control mode under the clinical setting.

Caption for Movie S9. Cadaver experiment: Intelligent manipulation under different surgical phases for a solo-hysterectomy. We have integrated and evaluated our developed OSFA system in a cadaver experiment, where a senior surgeon used the system throughout different phases.